

Beadwork Design Tool Project Report

The development of a browser-based design application for Kutch bead-weaving: the problem it addresses, how it was built, the challenges resolved along the way, and the studio's plan to launch it as a subscription product.

1 • OVERVIEW

Executive summary

Morii Design works with artisan communities in Kutch, Gujarat, whose bead-weaving tradition produces densely patterned textile pieces. The studio designs new patterns for these artisans; each design must be delivered as a colour chart that an artisan can follow bead by bead at the loom.

Existing design software could not represent the craft accurately. Beads are not square, and the weave does not place them in a square grid — so designs prepared in Photoshop were consistently distorted when woven. In June 2026 the studio identified this as the central problem and began building a dedicated tool. Over approximately six weeks, the tool progressed from a measured geometric model of the weave to a complete, tested design application, deployed continuously to a live site and used on iPad with Apple Pencil.

The tool is now feature-complete for its first market release. Morii Design is preparing to launch it as the studio's **first digital product**, on a **subscription model**, with branding and launch preparation currently in progress. A further capability — converting a photograph directly into a bead pattern — has since moved from standalone prototype to a **shipped, integrated feature** of the editor

(Section 9). The same underlying engine — a drawing grid built from a craft's real geometry rather than square pixels — has also been carried into a second, personal exploration applying the same idea to cross-stitch (Section 10).

PROJECT Beadwork — a design tool for the 3-bead weave	ORGANISATION Morii Design (first digital product)	DEVELOPMENT PERIOD June - July 2026
PLATFORM Web application; optimised for iPad + Apple Pencil	STATUS Feature-complete; launch preparation under way	BUSINESS MODEL Subscription

2 · THE STUDIO

About Morii Design

Morii Design is a textile design studio creating contemporary textile art rooted in traditional Indian craft. The studio designs in-house and makes with artisan partners: patterns are conceived by Morii's designers, then brought to life by craftspeople whose techniques carry generations of skill — among them the bead-weavers of Kutch, Gujarat, whose work this project serves. Each piece is both a work of contemporary design and a channel of artisan livelihood.

The name comes from *morii* — forest: a system of many living parts growing together. The idea runs through the studio's work, where individual stitches and beads accumulate into larger living surfaces, and it appears in this product too — every new artwork in the tool is automatically named after a tree (Oak, Neem, Willow, Cedar) until its designer names it.

The studio's working values — precision, fidelity between the designed and the made object, and respect for the craft traditions it draws on — defined this project's requirements. The tool exists because the studio was unwilling to accept a gap between what a designer approves on screen and what an artisan weaves; closing that gap digitally is a direct extension of how Morii already works in fabric. With this launch, those values enter a new medium: **Beadwork is the studio's first product made of software.**

The craft and the studio's role

Kutch beadwork is built from thousands of small glass beads woven into patterned fabric. The tool focuses on the **3-bead weave**, in which one upright "apex" bead sits above two tilted "base" beads; the unit repeats into a honeycomb-like structure. Morii Design's role is to create contemporary patterns for this technique and hand them to artisans in a form that can be woven exactly as designed.



Figure 1. A traditional Kutch beadwork piece. Each motif is composed of hundreds of individual beads.



Figure 2. The 3-bead weave in close-up: apex beads over pairs of tilted base beads, repeating into fabric.

4 • PROBLEM IDENTIFICATION

Why existing software failed the craft

Before this project, the studio's designers worked in two ways: drawing each bead by hand, which was accurate but extremely slow, or drawing in Photoshop, which was fast but inaccurate. The inaccuracy has a precise cause:

- **Bead proportions.** A pixel is square; a bead is not. The beads used in this weave have a width-to-height ratio of approximately 4:5.

- **Weave geometry.** Beads do not sit in aligned rows and columns. They occupy a staggered, brick-offset lattice: each row nestles into the gaps of the row below, alternate rows carry half as many beads, and base beads tilt $\pm 45^\circ$.

A pattern drawn on a square grid therefore changes shape when woven — motifs compress vertically and shear sideways. The woven piece never matched the approved design, and neither the designer nor the client could judge the final appearance from the screen.

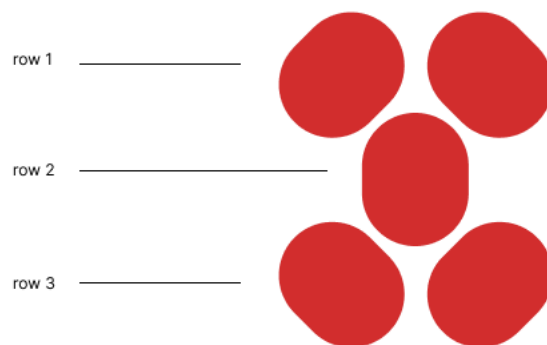


Figure 3. The studio's working sketch of the weave's row structure — horizontal apex rows alternating with mirrored tilted rows. This geometry became the specification for the tool's grid.

A drawing grid built from the real weave

The tool was conceived to bridge this gap directly: construct the drawing surface from the **measured geometry of the actual weave**, so that the screen, the printed chart, and the woven piece are three views of the same object.

Three principles were fixed at the outset and held throughout development:

- **Physical units everywhere.** The canvas is specified in centimetres and beads in millimetres; bead and row counts are derived from real dimensions. The lattice spacing was calibrated against a physical woven swatch — 36 beads across 7 cm at a 1.5 mm bead size.
- **One geometry, verified visually.** The same geometric model drives the on-screen editor and the printed chart, and every change to it was verified by overlaying renders on reference material rather than judged by eye.
- **The chart is the deliverable.** The end product of every design session is a print-ready chart the artisan reads at the loom: outlined beads, guide lines, edge numbering, and a colour key with exact bead counts.

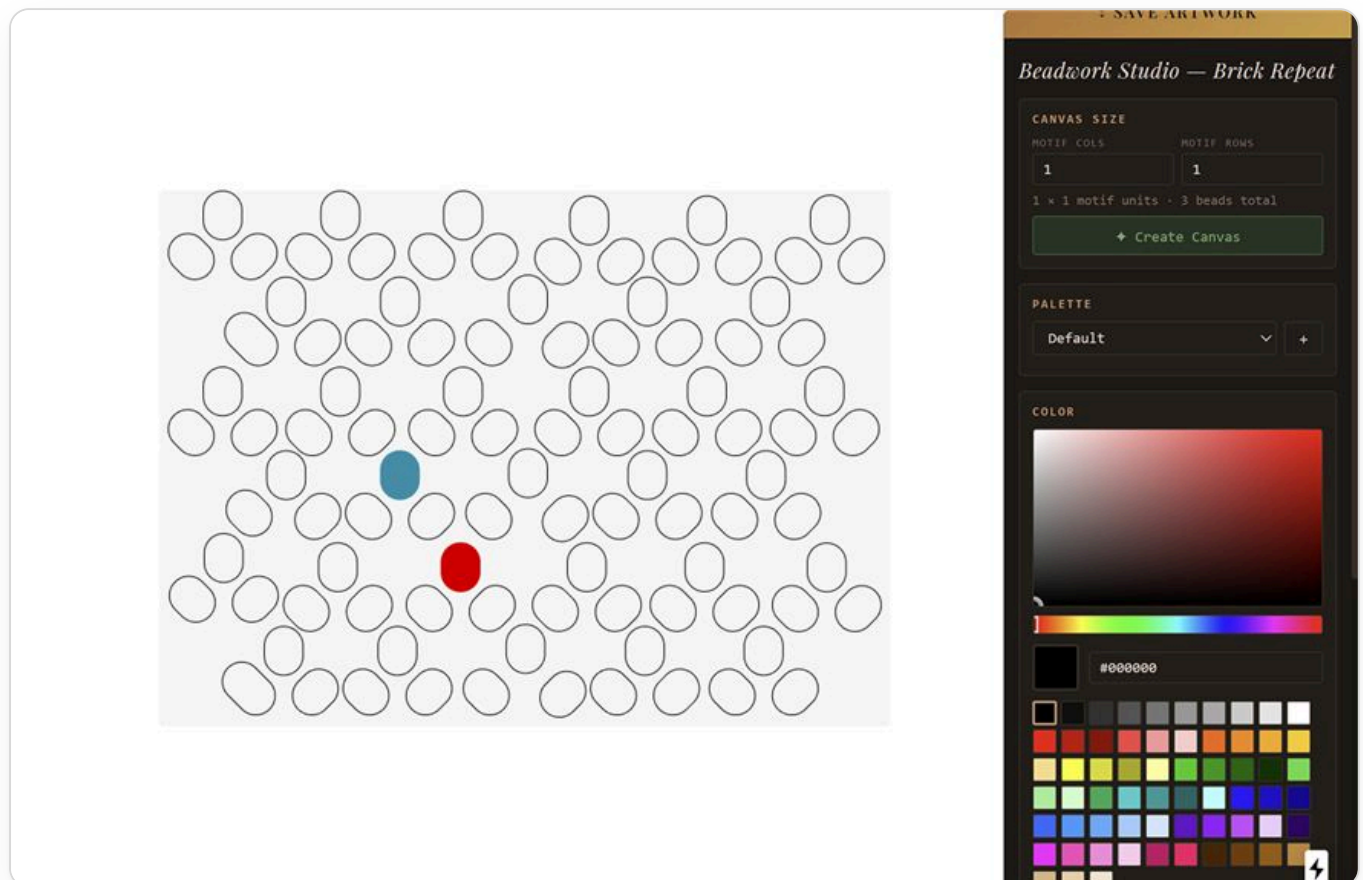


Figure 4. The original design mockup. The implemented grid was verified against this reference overlay by overlay.

Six weeks from model to product

Development ran in six phases, with close to ninety revisions deployed continuously to a live site. Each phase closed with automated browser tests verifying its features before release.

● PHASE 1 • EARLY JUNE 2026

Foundation: the geometric model

The application was rebuilt on a modern web stack (Vite, React 18). The lattice was measured from design vectors, calibrated against the physical swatch, and implemented as pure geometry separate from the interface — the part of the codebase everything else depends on.

● PHASE 2 • MID JUNE

Core design tools

Drawing, erasing and flood fill; colour palettes; marquee selection; a pattern maker with grid, brick and half-drop repeat layouts; mirroring; duplicate-and-place; layers; reference-image tracing; and a multi-artwork gallery with automatic saving. A second weave technique (1-bead loom) was added through a pluggable technique system.

● PHASE 3 • LATE JUNE

iPad platform work

Apple Pencil input with palm rejection, pinch zoom, two-finger rotation, and tap gestures for undo and redo. This phase also included a systematic stability investigation, described in Section 7, which took the application from frequent tablet crashes to sustained smooth operation on very large canvases.

● PHASE 4 • EARLY JULY

Interface redesign

The interface was redesigned in Figma and rebuilt as a focused workspace in the style of professional tablet drawing applications: floating toolbars, brush and palette rails, a layers panel, and light and dark themes, with the canvas occupying the full screen.

● PHASE 5 • MID JULY

Launch-readiness features

Shape assistance (hold-to-snap circles, lines, rectangles and polygons), a universal library of the studio's real bead colours, layer groups, a refreshed icon set, and the final rendering correction that made bead colour exact at every zoom level.

PHASE 6 · MID-LATE JULY

Photo import, gallery and touch usability

The photo-to-bead conversion prototype (Section 9) was integrated directly into the editor as an "Import photo as beads" flow, then refined so photos always open uncropped and reframing never silently discards work. The artwork gallery was rebuilt from a plain text list into a visual card grid with a live saved thumbnail per design, and the layers panel gained Procreate-style touch gestures — hold to reorder, double-tap to rename — matching how the studio's designers already work on iPad.

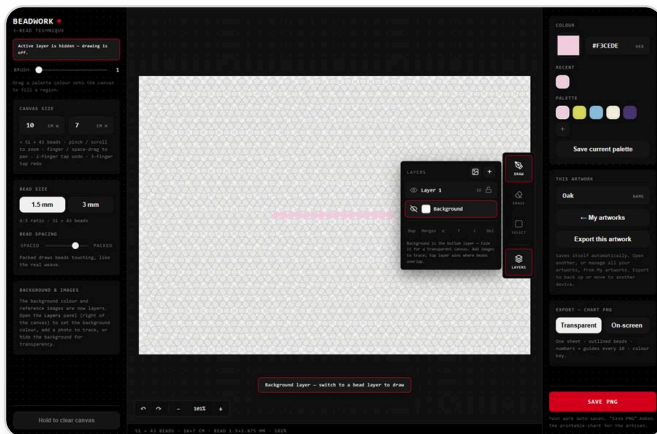


Figure 5. Phase 2 interface: complete in function, but the side panels competed with the canvas.

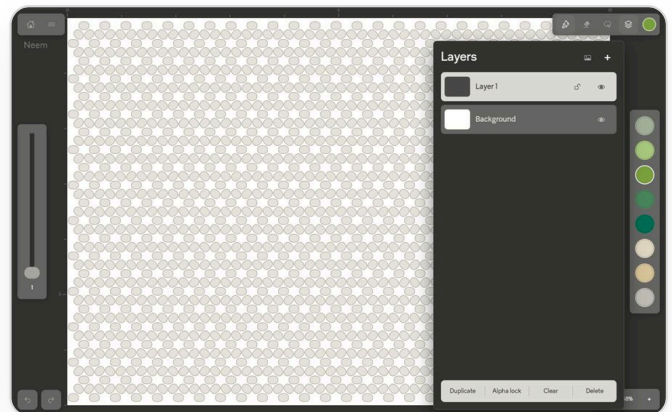


Figure 6. Phase 4–5 workspace: floating tools, layers panel, and the canvas as the primary surface.

7 · TECHNICAL CHALLENGES

Problems encountered and resolved

Four engineering problems materially shaped the project. Each was diagnosed with instrumentation, fixed, and covered by an automated regression test.

RESOLVED

7.1 · Application crashes on iPad

Safari on iPad terminated the application during drawing sessions. Diagnostic instrumentation built into the app — a crash recorder persisting the last known state — showed the cause was not memory, as first suspected, but **main-thread freezes**: every pencil movement (up to 240 per second) triggered a full repaint of every bead on the canvas. The rendering architecture was reworked so that strokes repaint only the beads they change, erasing reveals a pre-rendered underlayer, and completing a stroke no longer re-renders the document. The fix was validated by replaying recorded pencil sessions in an emulated Safari engine under 20× CPU throttling.

RESOLVED

7.2 · Rendering the woven appearance at scale

Rendering each bead as a detailed shape froze the application for seconds once designs reached tens of thousands of beads; rendering flat rectangles was fast but lost the woven appearance entirely. The solution renders colour as fast rectangles and overlays a single repeating bead-shaped stencil in one draw operation, independent of bead count. The woven texture is now visible at every zoom level on the largest supported canvas.

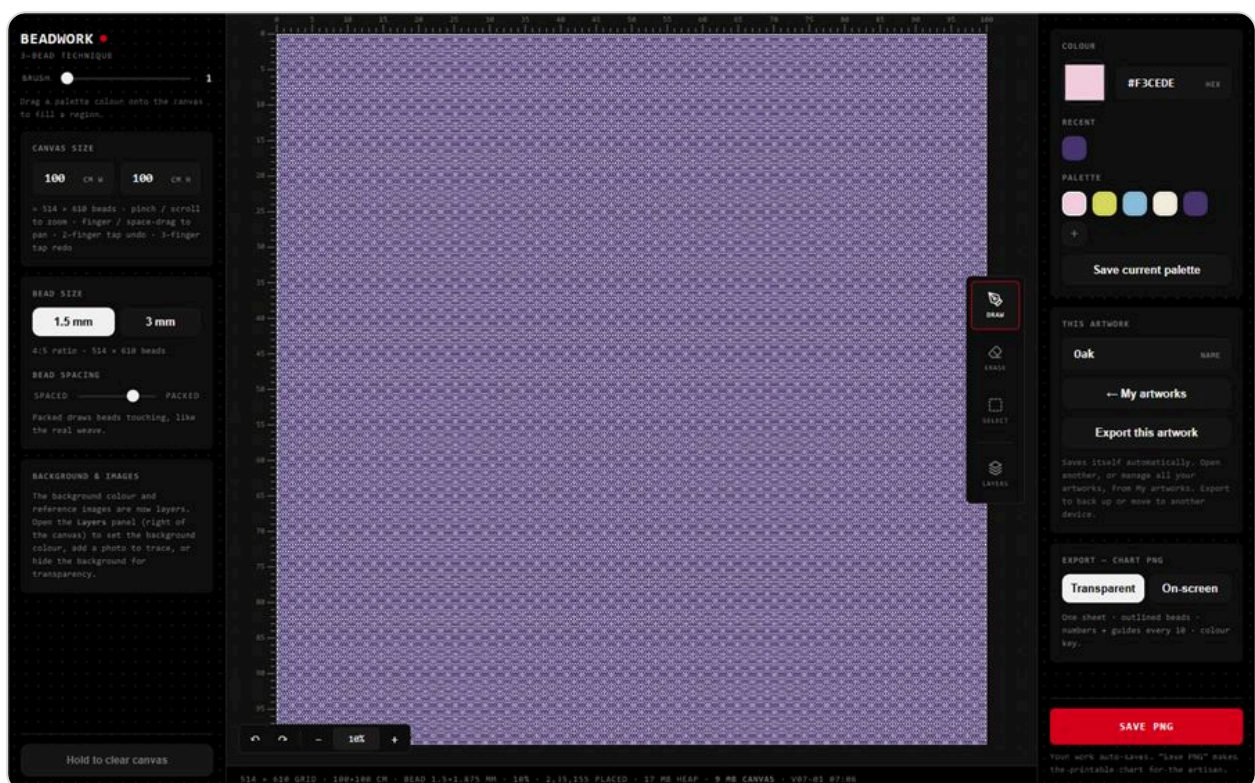


Figure 7. A fully filled 100 × 100 cm canvas — 235,155 beads — rendering with the woven texture at far zoom, without frame drops.

RESOLVED · JULY 2026

7.3 · Colour bleeding between beads

The stencil technique introduced a defect: the rectangle of colour behind each tilted bead extended past the bead itself, and fragments of it showed through neighbouring beads' stencil openings, so painted beads appeared to stain adjacent empty ones. An initial correction constrained each bead's colour to its bounding box; testing on the device found this still left faint fragments on tilted beads, so a second, more precise correction rotates each bead's colour patch to match its silhouette exactly. The same defect, present on the tool's second weave technique for a related reason, was found and resolved the same way. The automated test suite now checks for stray colour fragments around isolated beads on both techniques: before the correction, clearly visible slivers; after it, at most three pixels of anti-aliasing.

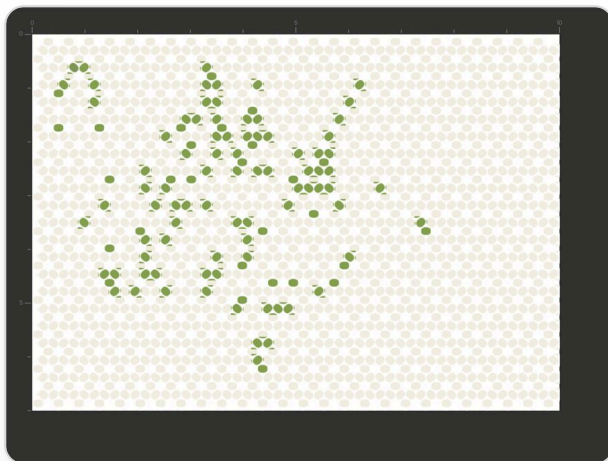


Figure 8. The defect as reported on iPad: colour fragments on neighbouring empty beads.



Figure 9. After the correction: every bead renders whole and separate at the same zoom.

RESOLVED

7.4 · Export reliability and format correctness

Browsers impose an undocumented maximum image size; charts exceeding it exported as blank files with no error, and large charts took over 45 seconds to

render, appearing frozen. The exporter now sizes its output to each browser's safe limit and renders in small batches that yield control back to the interface. A fully filled 40 × 40 cm chart (37,698 beads) exports in approximately 3.6 seconds with the application responsive throughout.

A related defect was found while preparing charts for artisan hand-off: PNG export always painted a white sheet behind the design, defeating the purpose of a transparent file meant to be placed over other material. PNG export now produces a genuinely transparent image; JPG, a format with no transparency channel, keeps the white sheet and the artwork's background colour for a print-ready look.

235,155

beads on the largest
canvas, rendered
smoothly

45 → 0

freeze events during
an erase stroke

12×

faster chart export
(45 s → 3.6 s)

60+

automated browser
checks per release

Every feature is covered by a scripted browser test that operates the real application — drawing, gesturing, exporting — and measures rendered pixels and main-thread freezes. Geometry and behaviour are verified against reference material, not assumed.

8 · THE PRODUCT TODAY

Feature summary at launch readiness

- **True-to-craft canvas** — sized in centimetres, bead counts derived from bead millimetres; two weave techniques (3-bead and 1-bead loom), selected per artwork.
- **Professional drawing tools** — draw, erase, flood fill, brush sizes, straight-line snapping, and shape assistance: holding the pen still snaps a rough stroke to an exact circle, line, rectangle or polygon.
- **Layers and groups** — visibility, locking, reordering, merging, alpha lock, collapsible groups, and reference-photo tracing with adjustable opacity.

- **Pattern tools** — motif repetition in grid, brick and half-drop layouts; selection mirroring for symmetric designs.
- **Colour management** — per-artwork palettes, a device-wide library of the studio's real bead colours, and a deliberately neutral interface that does not bias colour judgement.
- **Artwork management** — a gallery of artworks shown as a card grid with a live saved thumbnail per design, automatic saving, long-press/right-click rename · duplicate · delete, and file export/import for backup and transfer.
- **Import photo as beads** — a photograph opens uncropped, can be panned and reframed without losing work, and commits as an editable layer of beads in the chosen palette — see Section 9.
- **Chart export** — a print-ready chart with outlined beads, guide lines every ten rows, edge numbering, and a colour key with per-colour bead counts.

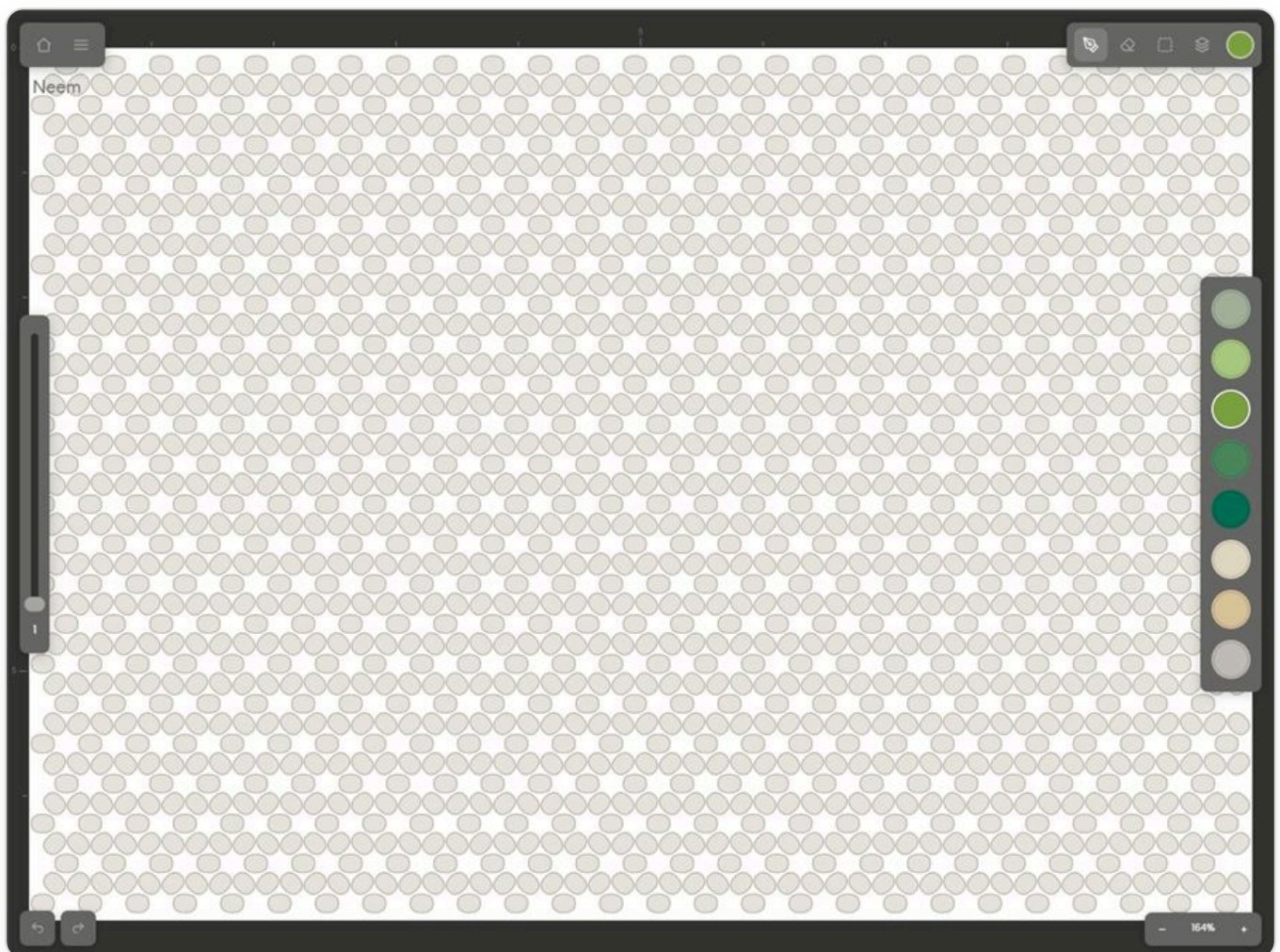


Figure 10. The editor at launch readiness. Interface chrome is kept neutral; the canvas carries the colour.

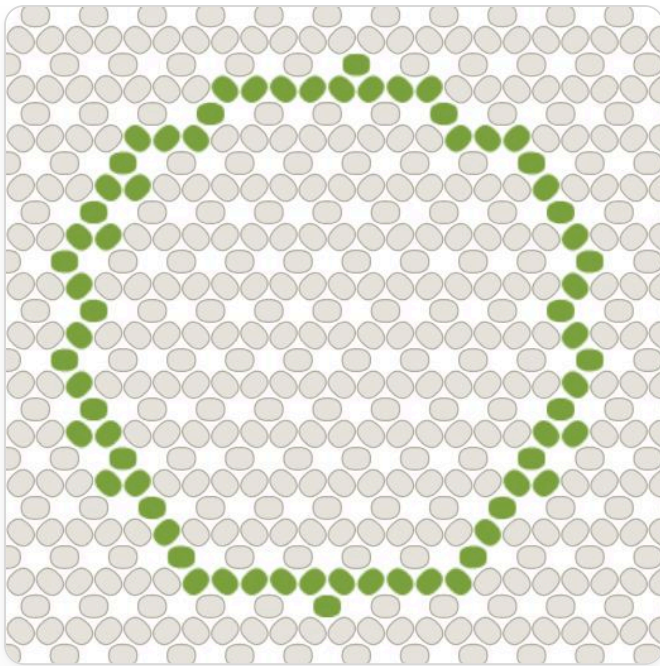


Figure 11. Shape assistance: a hand-drawn ring snapped to an exact bead circle.

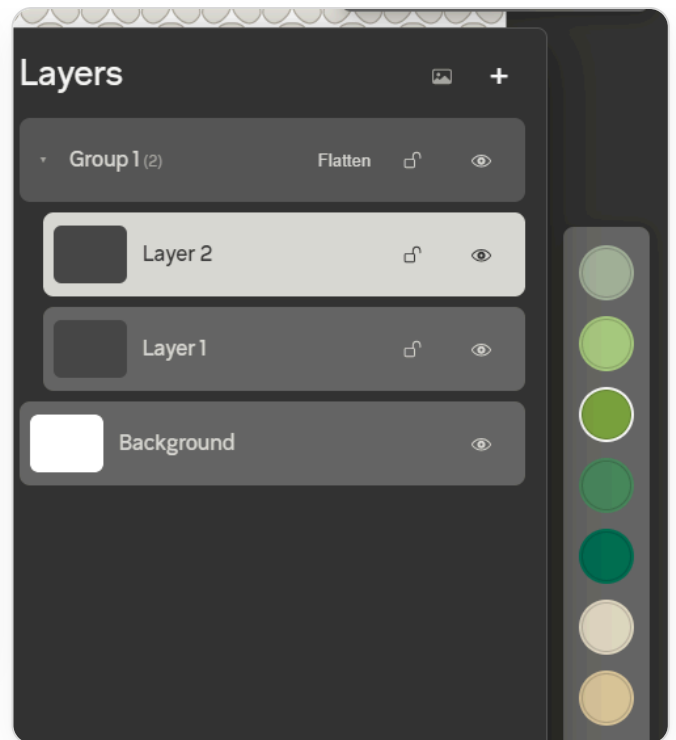


Figure 12. Layer groups with shared visibility, locking and flattening.

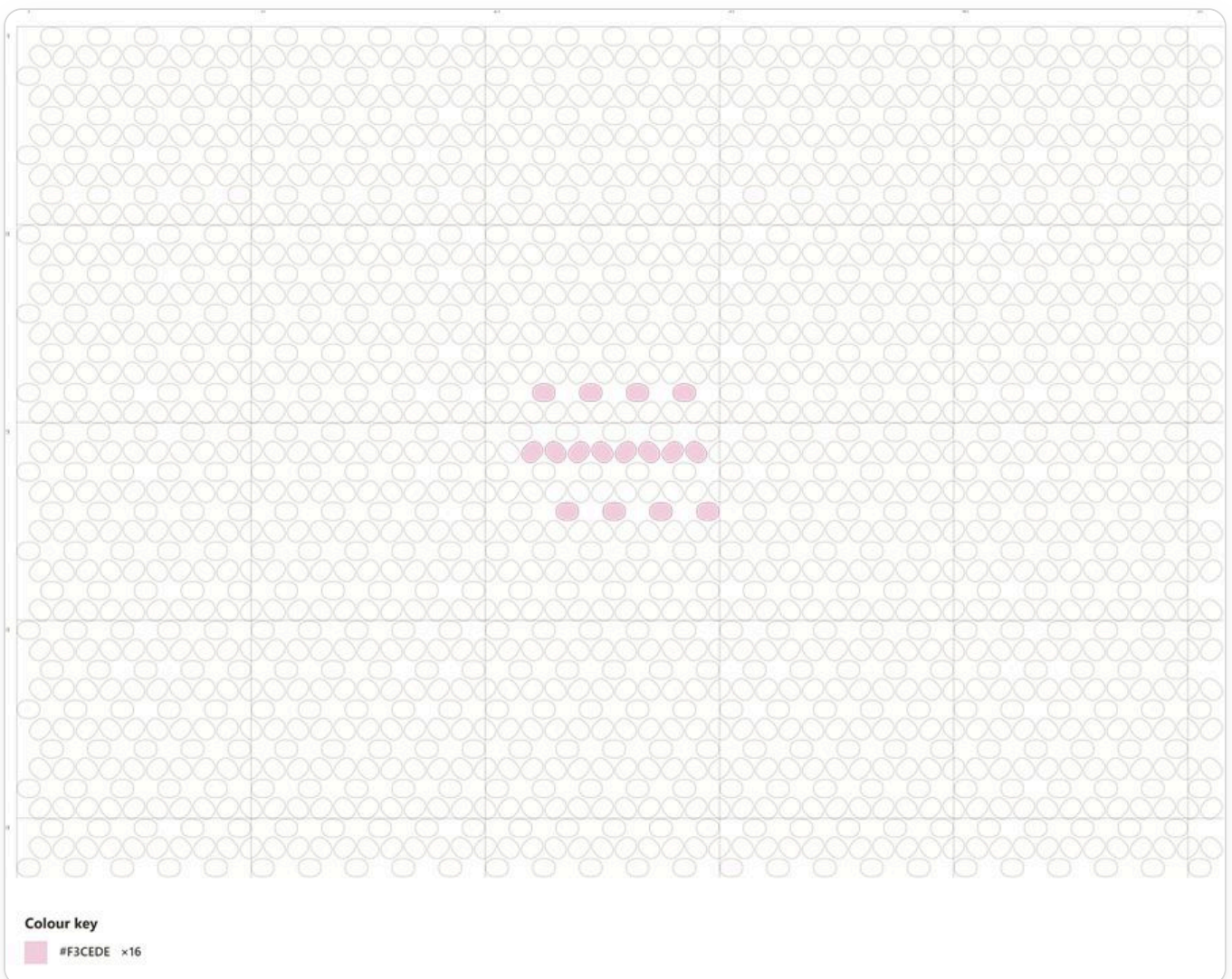


Figure 13. The deliverable: the artisan's chart, with outlined beads, numbered guides, and per-colour bead counts for ordering stock.

Design principle: neutrality toward colour

One interface rule has been enforced since the first specification: **the interface must not bias the designer's perception of bead colour**. Designers spend long sessions selecting bead colours against the canvas; saturated interface colours would distort those judgements. State is therefore communicated through tone and weight rather than hue, and the canvas remains the most colourful element on screen. This restraint is also a brand position: the design recedes so the craft can be judged accurately.

9 · SHIPPED FEATURE

Converting a photograph directly into a bead pattern

A capability first built as a standalone prototype — converting an uploaded photograph directly into a bead pattern, matched to a chosen colour palette and rendered on the true weave lattice rather than a plain pixel grid — has since been **integrated into the main editor** as an "Import photo as beads" flow, and the separate prototype was retired once its engine moved in. Each conversion commits as one bead layer per colour, grouped together, alongside the source photo kept as a hidden reference layer — a single undo step.

Two rounds of refinement followed integration, both driven by the same principle that governs the rest of the tool: never lose the designer's work silently. Photos now always open at full-fit rather than being silently cropped to the canvas; tapping the thumbnail opens a dedicated crop mode (pan, zoom, Fit · Fill · Done) instead of committing a guess; and the import modal was fixed so an accidental tap outside it can no longer discard an in-progress framing — Cancel is the only way out.

The conversion samples the source photograph once at every real bead position on the lattice, rather than treating the image as a plain grid of square pixels, so the result already respects the weave's staggered, half-density structure before any colour matching happens. Each sample is matched to the nearest colour in the working palette; an optional dithering step, adapted from a classic printing technique to the weave's honeycomb adjacency, preserves smooth gradients that a small palette would otherwise flatten into visible bands. The example below shows the same photograph converted with and without dithering, at 9,600 beads.

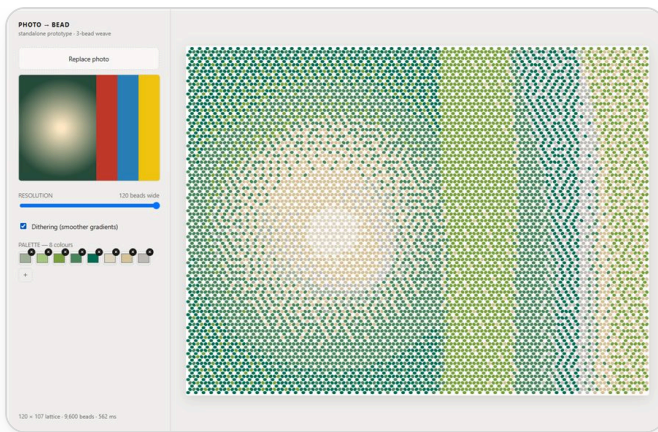


Figure 14. Converted with dithering: gradient areas of the source photograph read as smooth tonal transitions.

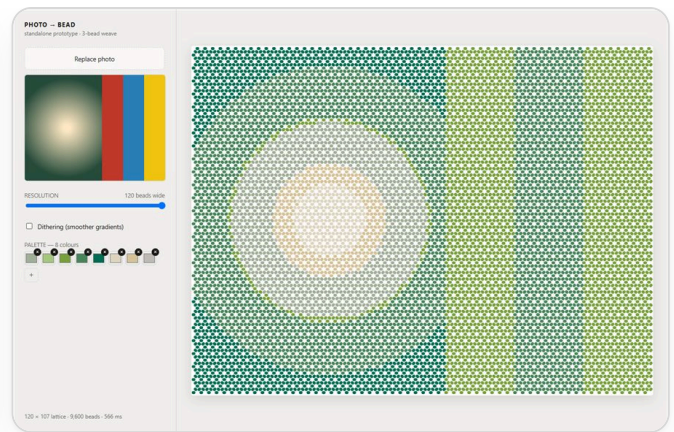


Figure 15. The same photograph without dithering: a flatter, more graphic result with visible colour banding.

During development, a rendering performance defect was found and corrected — the same class of issue already resolved once in the main application's chart exporter (Section 7.4): accumulating too many beads into a single drawing operation before rendering degrades disproportionately as bead count grows. At the import flow's highest resolution setting, this took the interactive preview from a 6.9-second freeze to 117 milliseconds, keeping every adjustment to the palette or resolution responsive.

10 • RELATED EXPLORATION

Applying the same engine to cross-stitch

Alongside the Beadwork product, a personal, self-directed exploration tested how far the same idea generalises: cross-stitch is a different craft with the same underlying flaw in existing software — a stitch drawn on a square-pixel grid does not have the same proportions as a real stitch sewn on aida cloth. Starting from Beadwork's codebase, the project replaced the bead lattice with cross-stitch's grid and shapes, then rebuilt the interface as the same canvas-first, iPad-style workspace already proven in the main product, including its drag-and-drop three-way brush and drawing-lag fixes.

This work sits outside Morii Design's Beadwork deliverable — it is not part of the studio's product or launch plan — but it is included here because it validates the core architectural bet behind Beadwork: that separating the weave's real geometry from the drawing and rendering machinery makes the tool portable to a different craft, not just a different pattern.

Morii Design's first digital product

Beadwork will be **Morii Design's first digital product**. The studio's work has until now existed in thread, bead and fabric; this launch extends the same standards — precision, respect for the craft, and fidelity between design and finished piece — into software built in-house.

The commercial model is **subscription-based**: the tool as an ongoing service for textile designers, craft studios and serious practitioners, with subscription revenue funding continued development. Launch preparation is currently in progress across four workstreams:

Brand identity

Naming, logo and typography derived from the weave's own geometry. Morii Lipi, the brand typeface, already leads the product and this report; a full branding brief has been prepared for the identity work.

Positioning

A professional instrument rooted in a real craft tradition — precise, calm and authentic, presenting Kutch heritage with respect rather than ornament.

Launch surface

A marketing site, pricing and plan structure, and onboarding for the first cohort of subscribing designers.

Service infrastructure

Subscriber accounts, secure payment handling, and cloud backup of artworks — the components that make a studio tool a dependable commercial service.

Roadmap beyond launch

- **Accounts and cloud synchronisation.** Artworks currently save on-device with file export for backup; the subscription launch requires sign-in, cloud storage and cross-device continuity.

- **Subscription infrastructure.** Plans, billing, trials, and licensing terms that keep the tool accessible to individual designers.
- **Additional weave techniques.** The grid engine is already modular (3-bead and 1-bead today); further stitches — 2-bead variants, netting, peyote-family weaves — can each be added as a technique module.
- **Print-grade chart output.** Multi-sheet A4/A3 PDF pagination with alignment marks for large pieces, alongside the current single-sheet chart.
- **Pattern library and community.** Shareable motifs and palettes, with the longer-term possibility of a marketplace connecting designers and artisan groups.
- **Collaboration.** Shared studio workspaces, review comments, and structured hand-off between design and production.
- **Presentation tooling.** A prototype already exists (the "Kinetic Lab") that imports a finished design and displays it as draping, physically simulated bead fabric for client presentations.
- **Artisan-facing features.** A chart-reading mode for use at the loom — large type, row-by-row progression, Gujarati localisation — completing the path from designer to weaver.

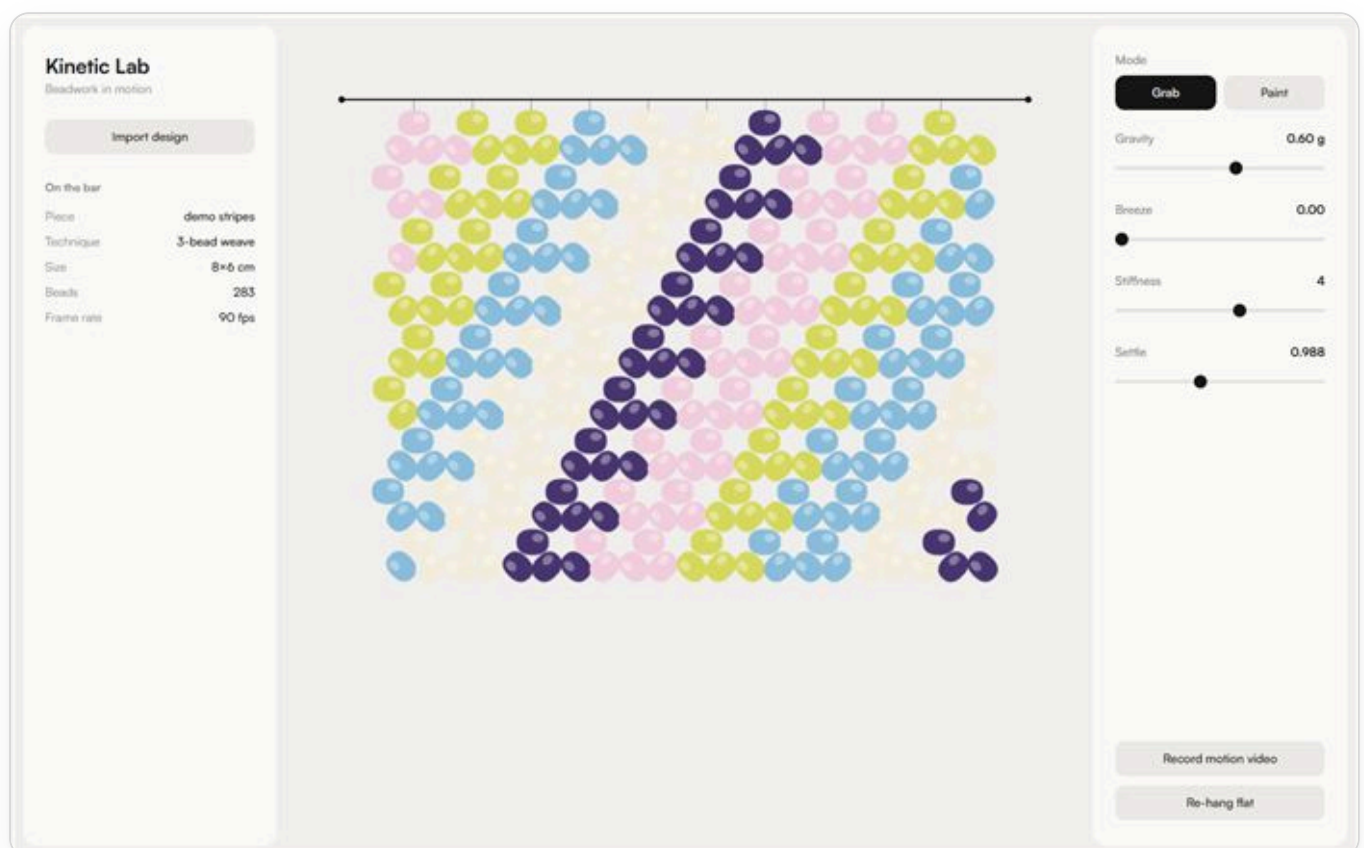


Figure 16. The Kinetic Lab prototype: a finished design rendered as draping bead fabric.

Assessment

In approximately six weeks, the studio identified a precise gap between digital design and a traditional craft, specified the geometry that closes it, and built a complete application on that foundation: accurate to the weave, fast enough for professional pencil work on a tablet, and verified by automated tests at every release.

The tool is ready for its market. The remaining work is commercial rather than technical — identity, pricing, accounts and onboarding — and is under way. When it launches, Beadwork will carry Morii Design into digital products for the first time, on the same principle the studio has always worked to: what is designed is what is made.

